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RESULTS

Results

01 · SIMPLE LINEAR REGRESSION

one numeric outcome, one predictor

Table 1. Regression coefficients

	(1)	β
(Intercept)	56.29 (1.42) [53.50, 59.08]	
study_hours_per_week	1.39 (0.20) [0.99, 1.78]	0.41
Num.Obs.	240	
R ²	0.17	
R ² Adj.	0.17	
F	48.59	
RMSE	7.41	
AIC	1648.70	
BIC	1659.14	
Log.Lik.	-821.35	

assumption checks: linearity, normality of residuals, homoscedasticity.

Figure. Fitted-line scatter

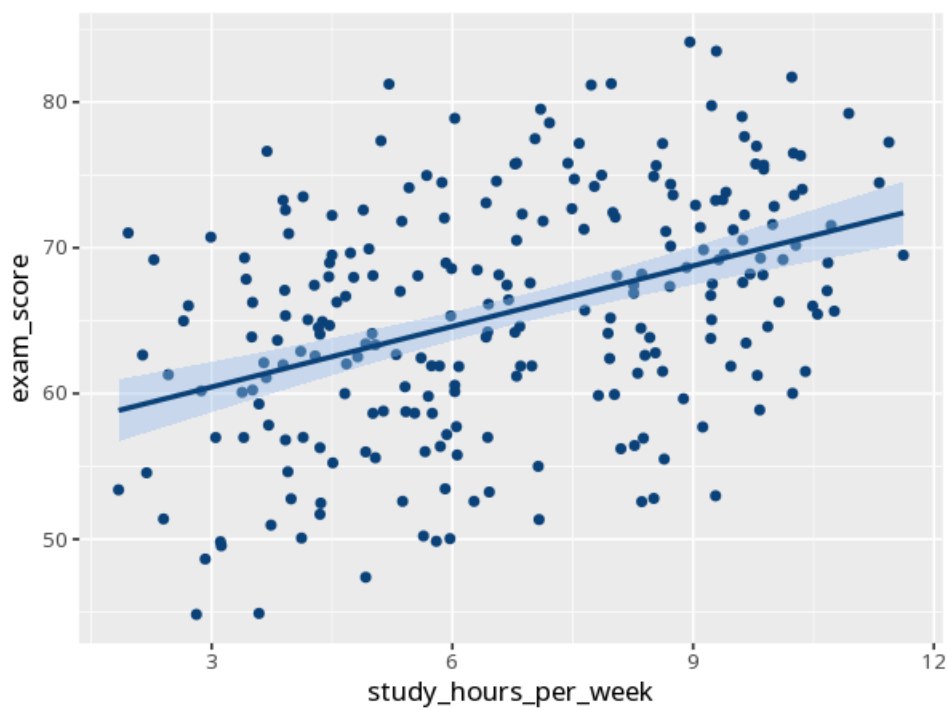
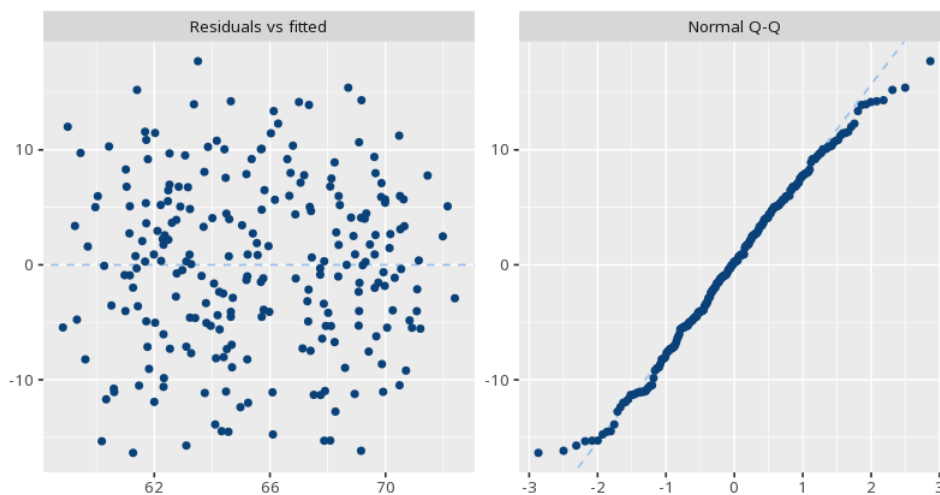


Figure. Residual diagnostics



Understanding the numbers

B. The slope of the line: the predicted change in the outcome for each one-unit increase in the predictor. Here, $B = 1.39$, so each one-unit increase in the predictor is associated with a change of 1.39 in the outcome.

β . The standardized slope: the change in the outcome, in standard deviations, per one SD change in the predictor - useful for comparing effect sizes on a common scale. Here, $\beta = .41$.

R^2 . The share of the outcome's variance explained by the predictor. Here, $R^2 = .17$.

R^2 Adj.. R^2 , adjusted for the number of predictors - the fairer figure when comparing models with a different number of predictors. Here, adjusted $R^2 = .17$.

F. The overall F-test of whether the model explains more variance than a model with no predictors at all. Here, $F = 48.59$.

RMSE. The root-mean-square error: the typical size of the model's prediction error, in the outcome's own units. Here, RMSE = 7.41.

AIC. The Akaike information criterion: a fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here, AIC = 1648.70.

BIC. The Bayesian information criterion: like AIC but penalizes complexity more heavily as sample size grows (lower is better, for models on the same data). Here, BIC = 1659.14.

Log.Lik.. The log-likelihood: how probable the observed data are under the fitted model (higher, i.e. less negative, is better); it underlies AIC and BIC. Here, Log.Lik. = -821.35.

N. The number of observations (rows of complete data) the model was fit on. Here, N = 240.

How to read this test

R^2 is the share of outcome variance explained. The predictor's B is the slope (change in outcome per unit), with p testing whether it differs from zero and the CI showing precision. β is the standardized slope (change in outcome in SDs per 1 SD change in the predictor), useful for comparing effect sizes on a common scale. Your significance threshold (α) is 0.05.

APA template: A simple linear regression gave $B=1.39$, $t(238)=6.97$, $p < .001$, $R^2=.17$.

Statistical basis: Ordinary least squares regression coefficient (B), SE, CI (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Standardized coefficient (β) (Lüdtke D, Ben-Shachar MS, Patil I, Makowski D (2020). "Extracting, Computing and Exploring the Parameters of Statistical Models using R." Journal of Open Source Software, 5(53), 2445.). Full references in the exported CITATIONS.txt.

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